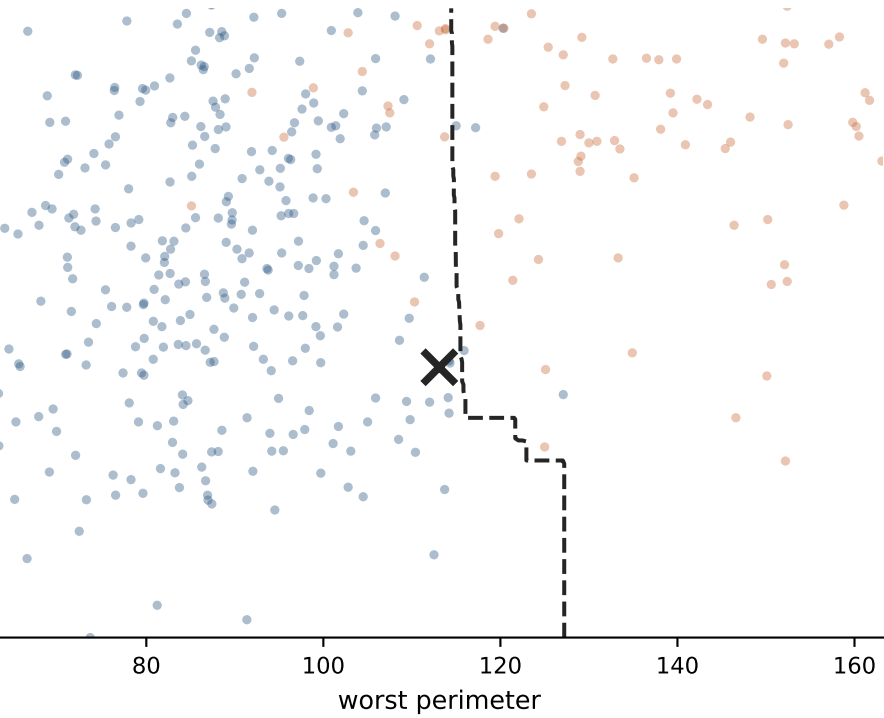


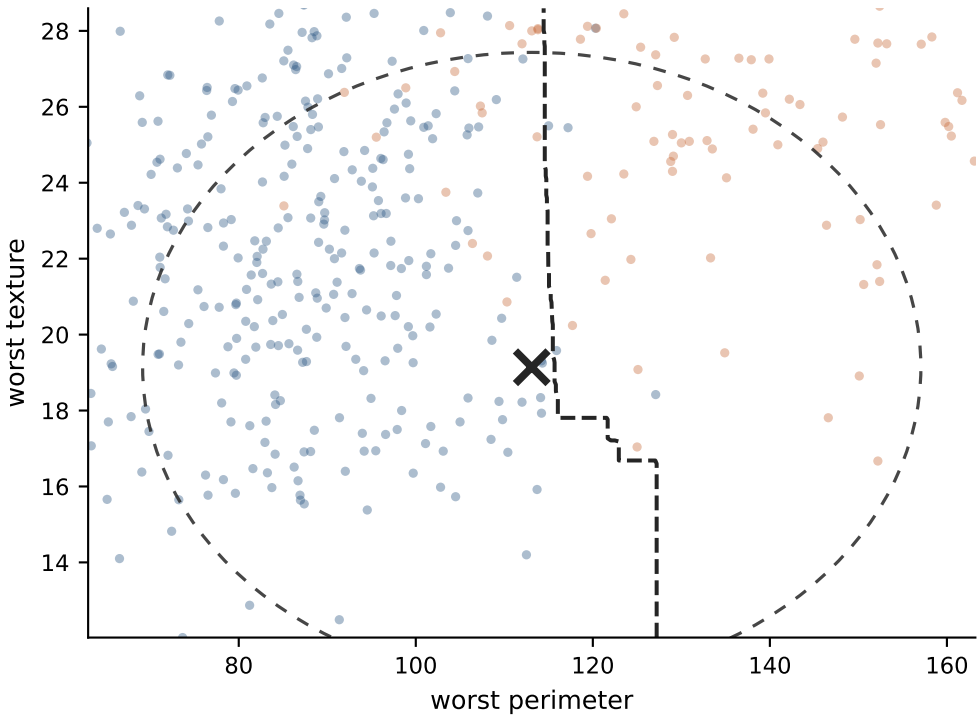
LIME

Local Interpretable Model-agnostic Explanation — patient #67, $P(\text{benign})=0.581$, $R^2=0.36$

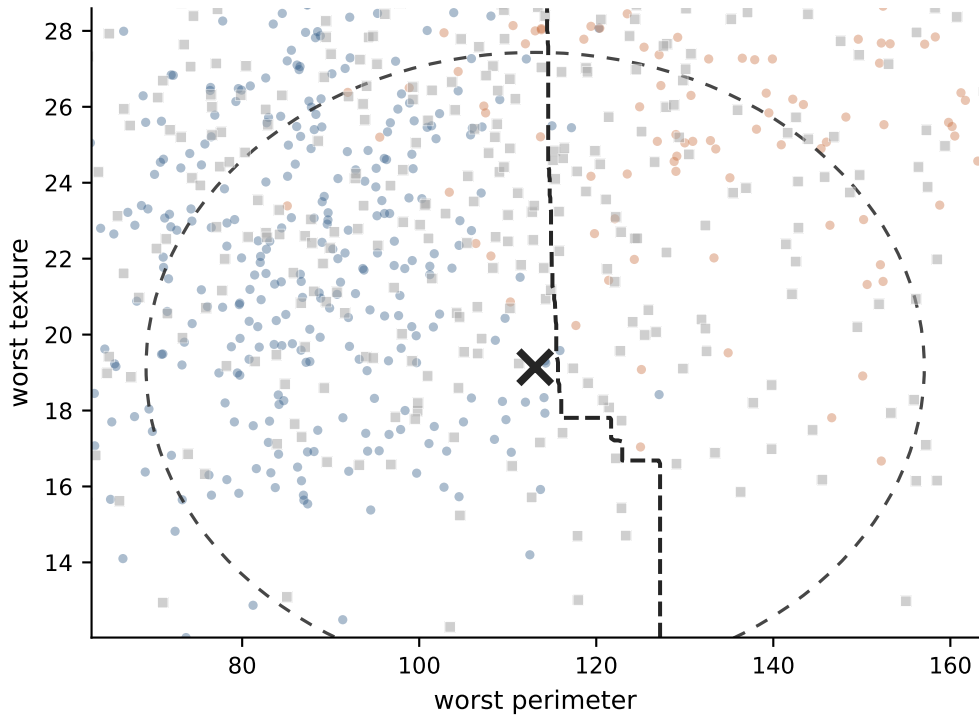
A · the black-box model



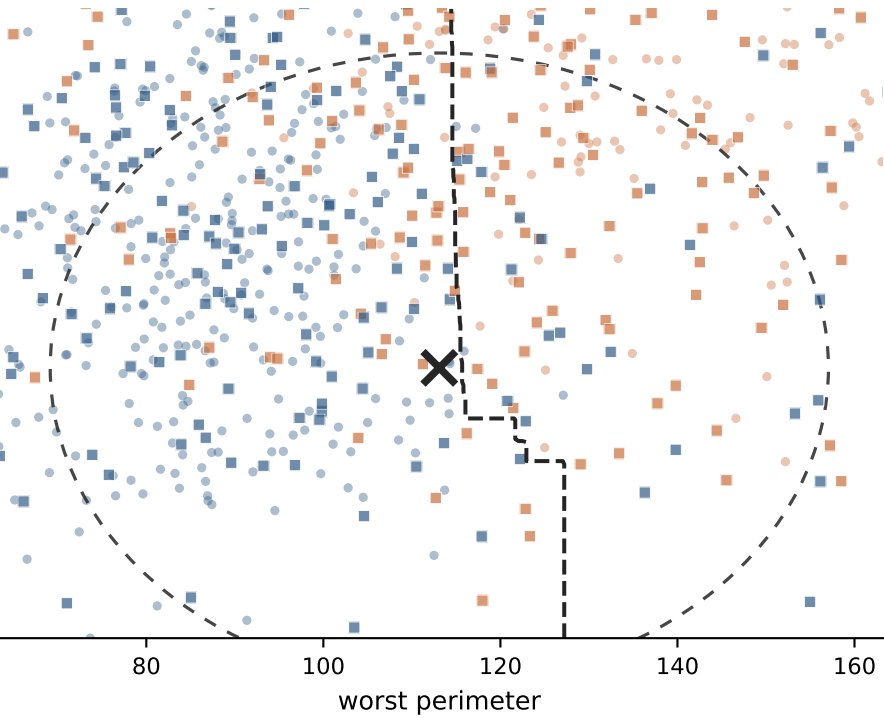
B · the local neighborhood



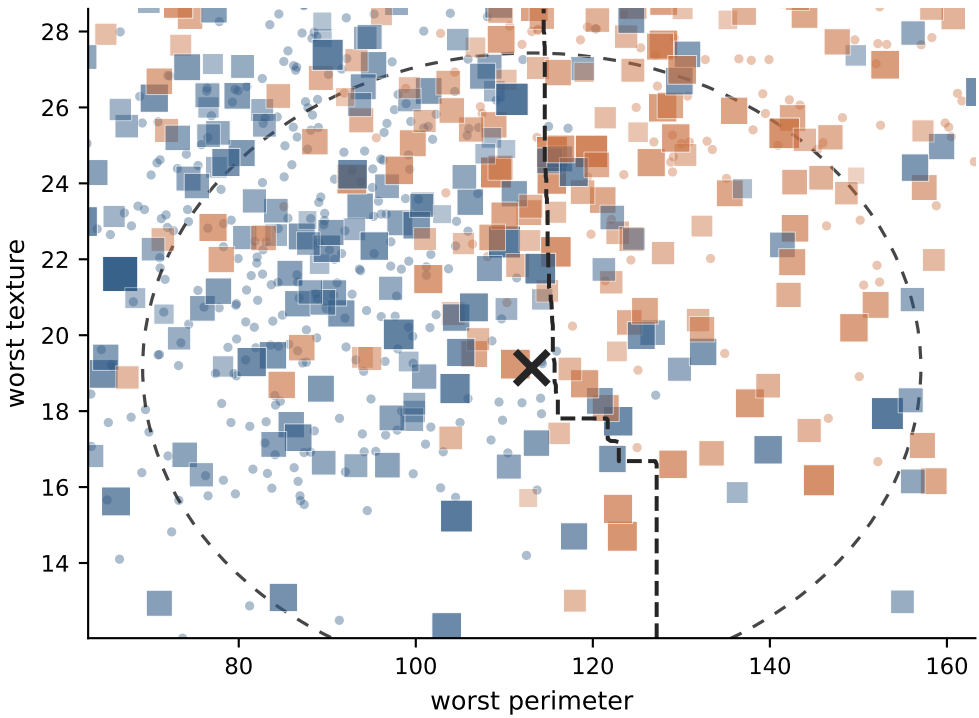
C · perturbation



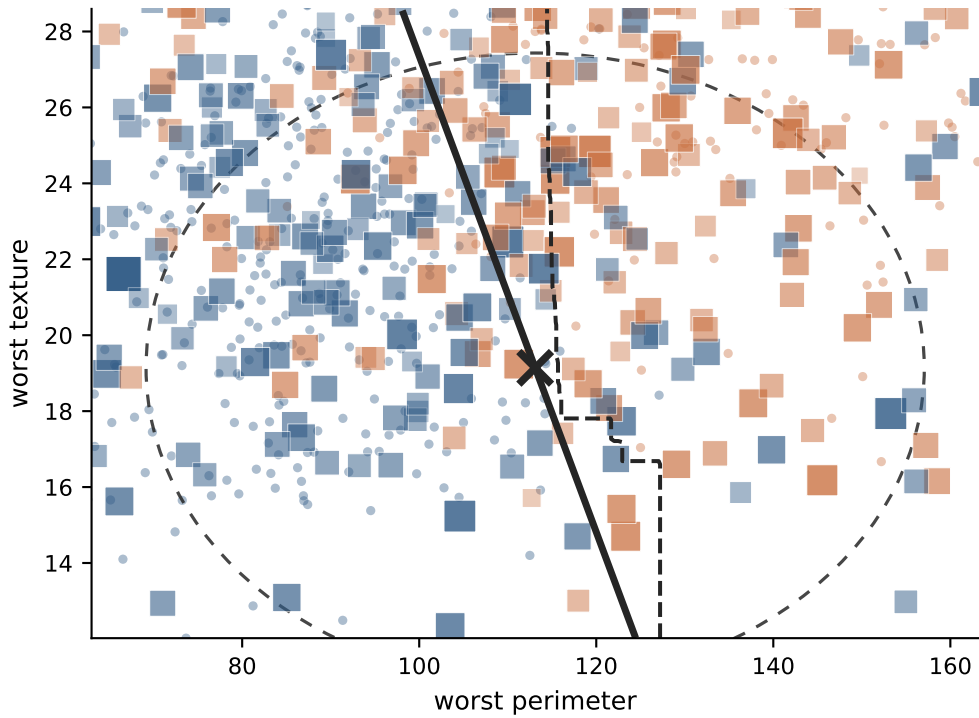
D · predictions on the neighbors



E · proximity weighting



F · the local fit



● real patient · benign
● real patient · malignant

■ neighbor · predicted benign
■ neighbor · predicted malignant

--- model boundary ($P=0.5$)
--- kernel weight = 0.95

✕ patient #67
— local fit (through the patient)